ÁRKTALAŠ HOLVVAS

ARCTIC SCREAM



VISUAL MANUAL · 1.5.3
FOR THE M-VAVE FM-1

Dark ambient drones and feedback screams from the arctic tundra.
Unofficial firmware by PolarPatch, free and open source (GPL-3.0); not made or endorsed by M-VAVE. Install at your own risk.

00

How to read this manual

Every page of Arctic Scream is shown as the FM-1's screen shows it, taken from the firmware itself, with a table of what its knobs and buttons do. The four tiles along the bottom of the screen always belong to KNOB 1–4, left to right, each in its colour.

Words used here

- **Tap** a button: open its page. Tap again: its next page.
- **Hold** a button: its second job (written as *hold*).
- **A + B**: hold A, then press B.
- **Keys from F2:** everything sits an octave down, for drones. **White keys 1–8** are counted from the left (F2, G2, A2, B2, C3, D3, E3, F3); the black keys from **F#2**.
- **T1–T4:** the four tracks. Turn ALGORITHM to pick one; most pages show the selected track.
- NEW marks what is new since 1.5, up to 1.5.3.

Arctic Scream p0larpatch

by PolarPatch



A drone firmware for the M-VAVE FM-1: four tracks, nine sources, long dark effects, a 64-step sequencer, chords, an arpeggiator and song mode, with its own feedback voice, Holvut, that wakes on its own and screams, and JIEHKKI, a whole piece it plays by itself.

Free and open source (GPL-3.0). Try it in your browser first: the demo, the editor and the cheat sheet are at polarpatch.github.io/Arctic-scream-FM-1.

01

Getting around

Four tracks play together and share six voices. The buttons open the pages, the four knobs set the four tiles at the bottom of the screen, and these controls work everywhere. Every track starts with ICE WRAITH: press a low key and let it howl.

Every page

No button needed

Control	What it does	Range
ALGORITHM	Picks the track. Every page shows that track.	T1–T4
PRESETS	Loads the next or previous sound into the track (factory first, then yours). On the sequencer pages: the tempo.	44 factory + 64 yours
SELECT	On HOME: the tempo. On the sequencer: the cursor.	20–300 BPM
MASTER	Volume.	
OCT– / OCT+	Octave down and up. In the save menus: back and confirm.	
Keys	Play the selected track, snapped to the SCALE (chord mode: its chord; ARP on: arpeggiated).	F2 and up
LATCH ON: keys	A key starts its note, the same key again stops it.	
PLAY	Starts and stops all four sequences (following a MIDI clock: at its next pulse).	
REC	Record on or off. On SONG: arms song recording.	
REC: hold ~1 s	Clears the selected track's pattern (undoable).	
EDIT + OCT– / OCT+	Undo / redo the last edit.	
SEL + OCT+	The selected track's LATCH on or off, from any page (off lets go of its latched notes).	
Key lights	Latched and arpeggiated notes stay lit, and their lights follow OCT– and OCT+.	
OCT– + OCT+	Sets SYSTEM 2's CPU PEAK and USB REC DROPS to 0.	

> Saving is automatic: stopped and silent, your work saves itself and comes back at power-on. Under a latched drone it saves after 30 s left alone; a slot or a sound you save is written at once (a short gap in the sound). Knobs speed up: slowly for fine steps, fast to sweep.

Buttons: tap and hold

What each button opens

HOME
HOME → MIXER 1 → MIXER 2. Hold: SYSTEM (then HOME: SYSTEM 2, SYSTEM 3).

EDIT
SOUND → SOURCE (FM: SOUND → FM ALGO → FM OP → SOURCE; Holvut: SOUND → LOOP → SOURCE).

ENV
ENV.

LFO
LFO → DRONE → JIEHKKI → RUOHČAT.

SEL
SCALE → CHORD. Hold and let go: chord mode on or off.

ARP
ARP. Hold: the arpeggiator on or off for the track.

SEQ
GRID → PATTERN SETTINGS. Hold + white keys: sequencer tools; hold + SELECT: pattern length.

SAVE
SOUNDS → PROJECTS. Hold + white keys 1–4: recall slots A–D, 5–8: save into them.

GLO
SONG. Hold + white keys 1–4: mute T1–T4, 5–8: solo them.

FX
TAPE → GRAIN → DELAY → SPACE → MASTER → DRIVE → AIR. Hold: PERFORM (+ black keys: GENERATE).

02

HOME and the mixers

Each track's wave glows on HOME as it plays, under a starry sky. Tap HOME again for the mixers: MIXER 1 draws each track as a curtain of standing ribbons, MIXER 2 as a ribbon across the stereo field.

HOME

Tap HOME



Control	What it does	Range
KNOB 1 VOLUME	The selected track's level (MUTE when muted).	0–100 %
KNOB 2 CUTOFF	The track filter's cutoff (MIXER 2's TYPE), for every source.	~30 Hz – OPEN
KNOB 3 WARP	Eight layers of motion that come in one after another: the voices drift apart, the filter breathes, the knobs wander, the storm pushes, vibrato, resonance, wide drift, at the top flutter and a sagging pitch. Smooth from still to wild; the level held.	0–100 %
KNOB 4 MORPH	One knob through all four of the source's knobs at once (the stepped ones, such as MODEL, stay), on a path of its own, wandering a little around it. At 0 the sound is as set.	0–100 %
SELECT	The tempo (following a MIDI clock: EXT and its tempo).	20–300 BPM

MIXER 1

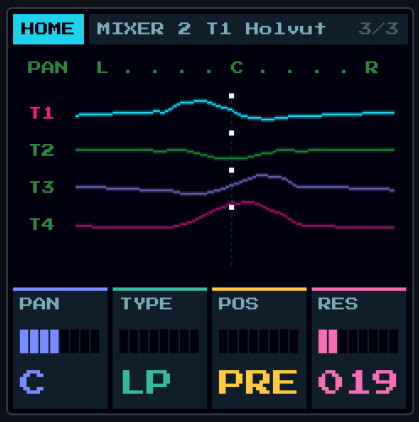
Tap HOME twice



Control	What it does	Range
KNOB 1 LEVEL	The track's level.	0–100 %
KNOB 2 DLY SEND	Send to DELAY.	0–100 %
KNOB 3 REV SEND	Send to SPACE.	0–100 %
KNOB 4 GRAIN SEND	Send to GRAIN.	0–100 %

MIXER 2

Tap HOME three times



Control	What it does	Range
KNOB 1 PAN	Where the track sits.	L64 – C – R63
KNOB 2 TYPE	The track filter: low-pass, high-pass or band-pass. LP open (CUTOFF at the top) or HP open (at the bottom): no filtering. BP is gentle at RESONANCE 0 and narrows as you turn it up.	LP / HP / BP
KNOB 3 POS	PRE: the effect sends hear the filtered sound. POST: only the dry sound is filtered; GRAIN, DELAY and SPACE hear it unfiltered (a dark track with a bright reverb).	PRE / POST
KNOB 4 RESONANCE	The track filter's resonance (on HOME before 1.3).	0–100 %

03

Sources: EDIT

EDIT opens the track's SOUND page, the source's four knobs, then SOURCE (a Holvut track has LOOP between them). Nine sources, each with its own picture.

SOURCE

Tap EDIT twice (Holvut: three times)



Control	What it does	Range
KNOB 1 SOURCE	Picks the track's source.	VA, FM, SAMPLE, VOICE, WHEEL, PHYS, PHASE, Holvut, Biegga
KNOB 2 LATCH	Holds the notes: a key starts its note, the same key again stops it.	OFF / ON

Holvut, the feedback voice

EDIT, source Holvut



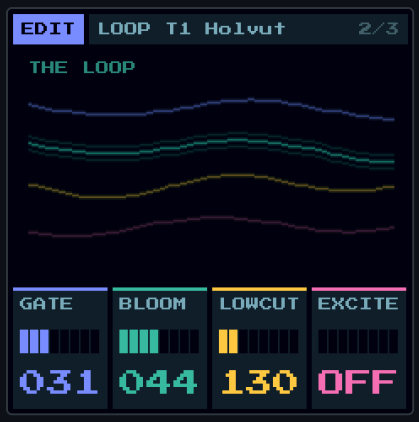
Control	What it does	Range
KNOB 1 FEEDBACK	Wakes the loop. Below the threshold it sleeps (THE LOOP: ASLEEP); past ~2/3 it grows out of silence and howls by itself (HOWLING). Its last steps (120–127) overblow: the howl leaps up the overtones.	0–100 %
KNOB 2 BODY	Bends it: the resonant body, ~1 to ~92 ms. Past 100 slow beating howls, past 115 slow pulses.	0–100 %
KNOB 3 DAMP	Turns the scream into a growl.	0–100 %
KNOB 4 DRIVE	Tears it.	0–100 %

> Holvut is North Sami for to howl. A string and a resonant body sit inside a loop with saturation, filters and diffusion; nothing is struck. Each key sends a burst into the loop, so a note near the threshold sets it off.

> Pressure: from a MIDI keyboard with aftertouch, pressing harder feeds the note back more, then overblows.

LOOP

EDIT twice on a Holvut track



Control	What it does	Range
KNOB 1 GATE	When the key is let go, the loop stops feeding itself: all the way up it dies into its own tail, in between it fades but lingers.	OFF, 1–100 %
KNOB 2 BLOOM	What GRAIN, DELAY and SPACE add to the mix goes back into the loop: an FX loop inside the feedback. Turn up the track's sends.	OFF, 1–100 %
KNOB 3 LOWCUT	A high-pass inside the loop, two poles: the howl thins and climbs; with DAMP, a band that screams.	~70 Hz – ~1.9 kHz
KNOB 4 EXCITE	The track's sample (SAMPLE's SLOT, START, LENGTH, looped) fed into the string: your own sound wakes the howl. The tile shows its slot (S2 45%) or EMPTY.	OFF, 1–100 %

> All four at OFF sound as before 1.4. Saved with the sound and the project.

> Each knob is a ribbon: GATE's tail fades, BLOOM's echoes appear around it, LOWCUT takes the swell out of it, EXCITE throws sparks along it.

VA

EDIT, source VA



Control	What it does	Range
KNOB 1 WAVE	Virtual analog: the waveform.	SAW / SQR / SUPER
KNOB 2 DETUNE	Detunes the voices against each other.	0–100 %
KNOB 3 CUTOFF	The source's own filter.	0–100 %
KNOB 4 RESONANCE	Its resonance.	0–100 %

FM: SOUND, FM ALGO, FM OP

EDIT on an FM track: SOUND → FM ALGO → FM OP



Control	What it does	Range
SOUND · KNOB 1	PATCH: the six-operator patch.	
SOUND · KNOB 2–4	BRIGHT, DECAY, DETUNE.	0–100 %
ALGO · KNOB 1	ALGO: how the six operators are wired.	01–32
ALGO · KNOB 2	FEEDBACK.	
ALGO · KNOB 3	OP: the operator the next page edits.	1–6
ALGO · KNOB 4	FINE: the operator's ratio, fine.	
OP · KNOB 1–4	RATIO, LEVEL, ATTACK, DECAY of the operator.	LEVEL etc. 0–99

> The screen draws the wiring: carriers (the ones you hear) outlined, modulators filled, the selected one lit.

SAMPLE

EDIT, source SAMPLE



Control	What it does	Range
KNOB 1 SLOT	One of the four sample slots, played as a loop.	1–4
KNOB 2 START	Where the loop starts.	0–100 %
KNOB 3 LENGTH	How long it is.	
KNOB 4 REVERSE	Backwards.	OFF / ON

> No samples are built in: the four slots are yours, filled with WAVs from the editor. An empty slot is silent.

VOICE

EDIT, source VOICE



Control	What it does	Range
KNOB 1 VOWEL	A singing formant voice: the vowel.	A E I O U
KNOB 2 SHIFT	Shifts the formants.	
KNOB 3 BUZZ	The buzz of the throat.	0–100 %
KNOB 4 BREATH	Air in it.	0–100 %

WHEEL

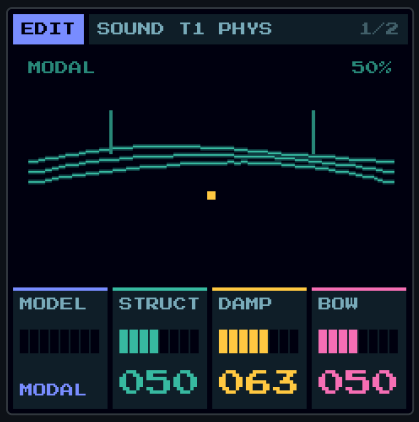
EDIT, source WHEEL



Control	What it does	Range
KNOB 1 DRAWBARS	A tonewheel organ: the registration.	16 settings
KNOB 2 PERC	Percussion.	OFF and 6 kinds
KNOB 3 CLICK	The key click.	0–100 %
KNOB 4 DRIVE	Overdrive.	0–100 %

PHYS

EDIT, source PHYS



Control	What it does	Range
KNOB 1 MODEL	Bowed and struck strings, bars, plates, membranes (MEMB), sympathetic strings (SYMP).	
KNOB 2 STRUCTURE	The body's structure (MEMB: HARM, SYMP: CHORD).	0–100 %
KNOB 3 DAMP	How fast it dies away.	0–100 %
KNOB 4 BOW	Bowing (MEMB: BEND, SYMP: BUZZ).	0–100 %

> The names of KNOB 2 and 4 change with the model; the screen shows the current one.

PHASE

EDIT, source PHASE



Control	What it does	Range
KNOB 1 WAVE	Phase distortion: the wave.	
KNOB 2 WAVE2	The second cycle's wave.	
KNOB 3 DCW	How far it is bent.	0–100 %
KNOB 4 DETUNE	Detune.	0–100 %

Biegga, the wind NEW

EDIT, source Biegga



Control	What it does	Range
KNOB 1 GUST	How hard and how often it gusts: steady at 0, every second or so at the top, the howl bending up as it blows.	0–100 %
KNOB 2 HOWL	The howl's share and how narrow it sings: a breath of rush at 0, a whistle at the top.	0–100 %
KNOB 3 TONE	The rush under it: a far storm … snow hissing past.	~150 Hz – ~10 kHz
KNOB 4 SNOW	Grains of snow ticking on the ice.	0–100 %

> Biegga is North Sami (and Lule Sami) for wind. Noise sings at the note like wind in a gap, so a chord of Biegga is a chord the wind sings. About as light on the processor as VA.

> The factory sound BIEGGA (TEXTURE) holds its note, gusts, and its storm rises over 4 bars.

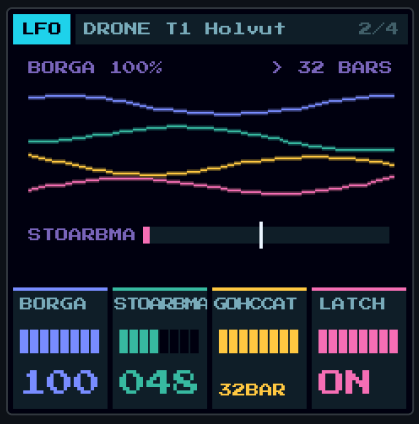
04

DRONE: a sound that lives on its own

Press LFO twice. DRONE makes a held sound move by itself, slowly, and never the same way twice. The knobs have North Sami names.

DRONE

Tap LFO twice



Control	What it does	Range
KNOB 1 **BORGA (snowdrift)**	Four slow random walks move the filter, the source's shape, its most natural knob (on Holvut: FEEDBACK, around the threshold) and the reverb's distance. Each voice drifts a few cents and breathes: a held chord sounds like a choir.	0–100 %
KNOB 2 **STOARBMA (storm)**	The tension: opens the filter, raises resonance and drive, spreads the voices, makes the walks deeper and faster.	0–100 %
KNOB 3 **GOHCCÁT (to wake)**	How long the storm takes to rise, in tempo. A drone woken from silence starts calm and builds.	OFF, 1–32 bars
KNOB 4 **LATCH**	Holds the notes.	OFF / ON

> At 0 % nothing is added. The settings are saved with the sound; the seven Holvut presets start with BORGA on.

JIEHKKI NEW

Tap LFO three times



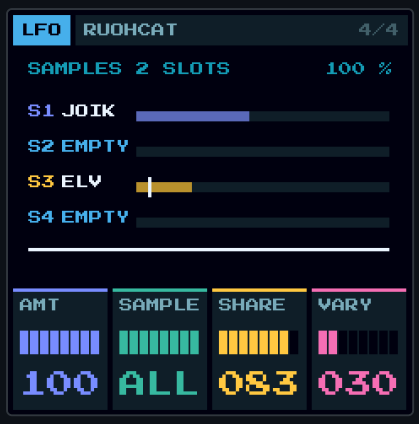
Control	What it does	Range
KNOB 1 **LENGTH**	How long the piece plays.	5, 10, 15, 25, 30, 60 MIN, ENDLESS
KNOB 2 **DRIFT**	How much moves: its own notes step through the scale, the root goes to its fourth or fifth and home, a track rests a while, the filters open over the first minutes and close as it dies out. 0: nothing moves, the piece only rises and dies out (a sleep timer).	0–100 %
KNOB 3 **RUOHČAT (to creak)**	The ice around the drone: cracks, drips and a deep boom now and then, from about one a minute to a storm. Plays whenever it is up.	0–100 %
KNOB 4 **START**	The piece. With nothing playing it plays its own notes, one a track from the SCALE, the tracks coming in one after another, heard from the first seconds; with notes playing it takes them and moves the mix.	OFF / ON

> Jiehkki is North Sami for glacier. Over its last fifth everything fades out; then the notes are let go, the sequencer stops, the screen goes dark and the FM-1 stays silent until you touch it.

> The glacier on the screen is the piece: its height the sound, the part played lit, now a white line. LENGTH, DRIFT and RUOHČAT are saved with the project.

RUOHČAT NEW

Tap LFO four times



Control	What it does	Range
KNOB 1 AMOUNT	RUOHČAT's knob on JIEHKKI, the same setting: how often.	0–100 %
KNOB 2 SAMPLE	Which sample slots it plays from (ALL: any that holds one).	OFF, SLOT 1–4, ALL
KNOB 3 SHARE	How many of its sounds are your samples. 100 %: only the samples.	0–100 %
KNOB 4 VARY	0: a sample whole, from its start, as recorded. Up: from further in, shorter, now and then a fourth, a fifth or an octave away.	0–100 %

Your own samples among the ice's sounds: a joik far off over the drone, water, wind. Load WAVs into the four slots with the editor (6.6 s a slot).

Two play at once at most, each fading in and out, panned, some of it in the reverb. SAMPLE, SHARE and VARY are saved with the project.

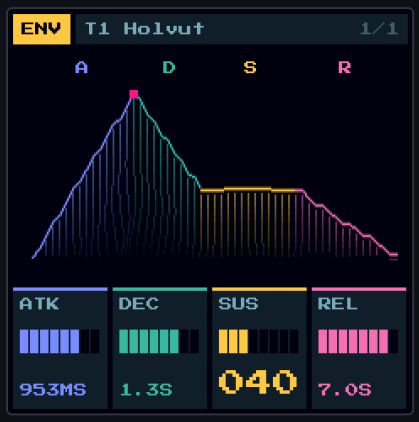
05

ENV and LFO

How each note starts and ends, and slow movement. Both belong to the track.

ENV

Tap ENV



Control	What it does	Range
KNOB 1 ATTACK	How each note starts.	~1 ms – 10 s
KNOB 2 DECAY	How fast it falls to SUSTAIN.	~10 ms – 10 s
KNOB 3 SUSTAIN	The level it holds.	0–100 %
KNOB 4 RELEASE	How it dies away after the key.	~10 ms – 20 s

LFO

Tap LFO



Control	What it does	Range
KNOB 1 RATE	How fast. It never restarts with a note, so drones keep breathing.	60 s a cycle – 20 Hz
KNOB 2 DEPTH	How much (0 = off).	0–100
KNOB 3 DESTINATION	What it moves. VOLUME swings around the track's VOLUME, so a track at 0 still sounds on its peaks: mute it with GLO + keys 1–4.	PITCH, VOLUME, PAN, the source's four knobs, REV SEND, GRAIN SEND, CUTOFF
KNOB 4 SHAPE	Its shape.	SINE, TRIANGLE, SAW UP, SAW DOWN, SQUARE, RANDOM, DRIFT

06

Scales, chords and the arpeggiator

SEL opens SCALE and CHORD; hold SEL and let go for chord mode: one key, a whole chord. ARP opens the arpeggiator.

SCALE

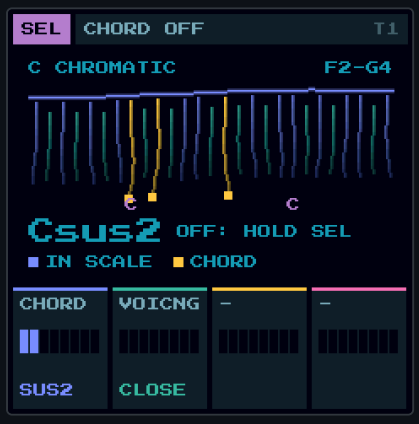
Tap SEL



Control	What it does	Range
KNOB 1 ROOT	The scale's root, shared by all tracks.	C–B
KNOB 2 SCALE	Keys, MIDI and recordings snap to it. The last is HORROR: close semitones, every note wrong in the right way.	OFF + 14 scales
KNOB 3 ŠUOKŅA NEW	The tuning from ROOT up, for every source (North Sami: a tone, its sound; SUOKNA on the screen). JUST: pure thirds and fifths, a held chord stands still. PYTHAG: pure fifths. HARMONIC: the harmonic series over ROOT, strange old colours. Saved with the project.	EQUAL, JUST, PYTHAG, HARMONIC

CHORD

Tap SEL twice



Control	What it does	Range
KNOB 1 CHORD TYPE	The chord a key plays in chord mode.	SCALE, MIN, MAJ, SUS2, SUS4, FIFTH, MIN7, MAJ7, DIM, ADD9, CLUSTER
KNOB 2 VOICING	How it is spread. Big chords use more of the six voices.	CLOSE, OPEN, SPREAD, +OCT, DROP

ARP

Tap ARP · hold ARP: on / off



Control	What it does	Range
KNOB 1 MODE	The order it plays the held keys in.	UP, DOWN, UP-DOWN, RANDOM, AS PLAYED, CHORD
KNOB 2 RATE	How fast, on the sequencer's clock.	1/1 – 1/32, triplets
KNOB 3 OCTAVES	How many octaves it climbs.	1–4
KNOB 4 GATE	How long each note sounds.	10–100 %

> It plays the keys you hold; with LATCH ON it keeps going after you let go.

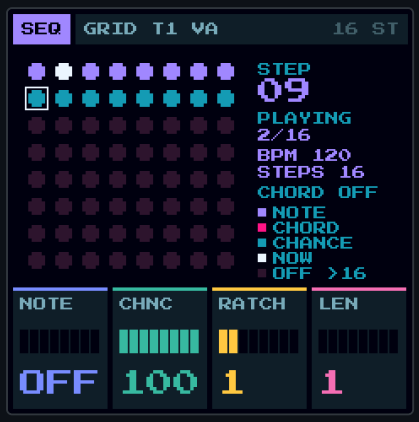
07

Sequencer: SEQ, PLAY, REC

Each track has its own 64-step pattern; all four play together on one tempo. Stopped + REC: each key fills a step and moves on. Playing + REC: notes land on the nearest step.

GRID

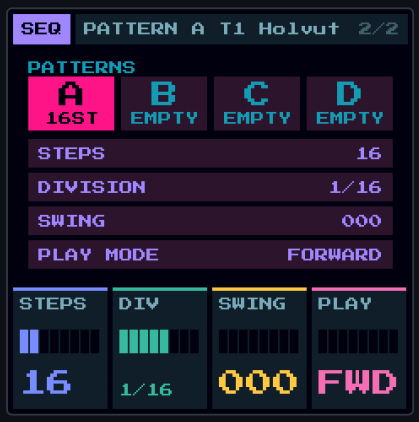
Tap SEQ



Control	What it does	Range
KNOB 1 NOTE	The selected step's note (all the way down: OFF).	OFF, notes
KNOB 2 CHANCE	How likely it plays.	0–100 %
KNOB 3 RATCHET	Repeats within the step.	1–4
KNOB 4 LENGTH	How long it sounds.	
SELECT	Moves the cursor. PRESETS: the tempo.	
SEL + a knob	Locks that value to the step. SEL + OCT–: clears the step's locks.	

PATTERN SETTINGS

Tap SEQ twice



Control	What it does	Range
KNOB 1 STEPS	The pattern's length (also: hold SEQ + turn SELECT).	1–64
KNOB 2 DIVISION	How long a step is.	1/16 etc.
KNOB 3 SWING	Swing.	0–100 %
KNOB 4 PLAY MODE	Which way it runs.	FORWARD and more

Hold SEQ + a white key (all undoable):

F2	G2	A2	B2	C3	D3	E3	F3
CLEAR	RANDOM	SHIFT ←	SHIFT →	COPY	PASTE	DOUBLE	MUTATE

08

Sounds, slots and songs: SAVE, GLO

A slot (A–D) holds the four tracks' sounds, sequences, locks and mix. Hold SAVE + white keys 1–4 to recall them (at the next bar while playing), 5–8 to save into them (press again within 3 s to overwrite).

SOUNDS

Tap SAVE



Control	What it does	Range
KNOB 1 SLOT	One of 64 slots for your own sounds.	001–064
KNOB 2 CATEGORY	Its category: it comes in PRESETS after that category's factory sounds.	
KNOB 3 TYPE	Shows only one source's sounds.	ALL, the sources
KNOB 4 ACTION	Confirm with OCT+, back out with OCT–.	LOAD / SAVE / CLEAR

A sound keeps its source and knobs, ENV, LFO, DRONE, LATCH, chord and (FM) its patch. Name it in the editor (up to 10 characters).

PROJECTS

Tap SAVE twice



Control	What it does	Range
KNOB 1 SLOT	One of the four slots.	A–D
KNOB 2 ACTION	Confirm with OCT+.	LOAD, SAVE, ERASE, NEW

SONG

Tap GLO



Control	What it does	Range
KNOB 1 SECTION	A step of the chain.	01–16
KNOB 2 PATTERN	The slot it plays (END: the song ends there).	A–D / END
KNOB 3 REPEATS	How many times.	01–16
KNOB 4 MODE	OFF: PLAY loops the slot. SONG: once. SONG LOOP: over and over.	OFF / SONG / LOOP

> Record a song: on SONG press REC, then PLAY, and switch slots with SAVE + keys 1–4.

> Hold GLO + white keys 1–4 mutes T1–T4, keys 5–8 solos them.

09

Destroy: FX

Tracks feed GRAIN, DELAY and SPACE through their sends; MASTER, DRIVE and TAPE work on the whole mix. Each page has its own northern lights.

TAPE

Tap FX

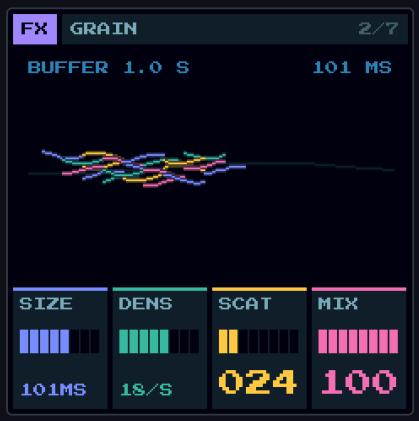


Control	What it does	Range
KNOB 1 WOW	Slow wobble and flutter, up to a seasick ±2 % of pitch.	0–100 %
KNOB 2 SATURATION	Tape saturation, up to 12x drive.	0–100 %
KNOB 3 TONE NEW	Darker as it turns up, down to ~400 Hz (DRIVE's TONE goes the other way).	OPEN, 10K … 400
KNOB 4 HISS	Tape hiss, up to loud.	0–100 %

All four at 0 switch TAPE off (BYPASS). It colours the whole mix at the very end.

GRAIN

Tap FX twice



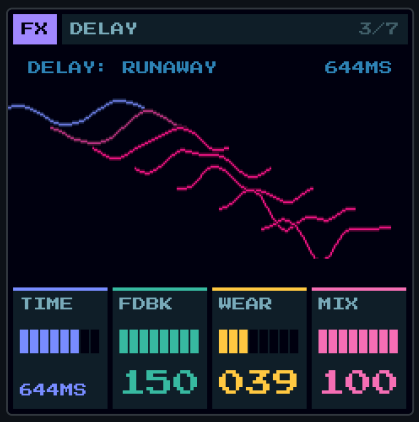
Control	What it does	Range
KNOB 1 SIZE	How long each grain is.	10–390 ms
KNOB 2 DENSITY	How many grains a second.	
KNOB 3 SCATTER	How far they scatter.	0–100 %
KNOB 4 MIX	How loud the cloud is.	0–100 %

FREEZE (PERFORM, F2) holds the grain cloud. GRAIN pours into SPACE.

GRAIN hears what the tracks send it (MIXER 1). Turn a knob up while the selected track sends nothing and its send goes to 50 %.

DELAY

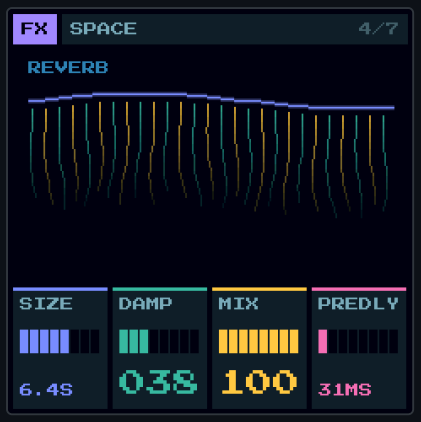
Tap FX three times



Control	What it does	Range
KNOB 1 TIME	The echo's time.	20 ms – 2.6 s
KNOB 2 FEEDBACK	Past 100 % the echo grows and saturates, and burns magenta on the screen: runaway.	0–150 %
KNOB 3 WEAR	Worn, wobbly echoes: darker, driven to 4x, wobbling up to ~11 ms.	0–100 %
KNOB 4 MIX	How loud they are.	0–100 %

SPACE

Tap FX four times



Control	What it does	Range
KNOB 1 SIZE	The reverb's length; at the top it holds for ever (INF, FREEZE).	to ~32 s, INF
KNOB 2 DAMP	Darkens the tail.	0–100 %
KNOB 3 MIX	How loud it is.	0–100 %
KNOB 4 PRE-DELAY	The gap before it.	0–250 ms

MASTER

Tap FX five times



Control	What it does	Range
KNOB 1 CUTOFF	A filter on everything, in stereo. LP at the top or HP at the bottom: off completely (OPEN).	~30 Hz – OPEN
KNOB 2 RESONANCE	Its peak at the cutoff: the rays there flare.	0–100 %
KNOB 3 TYPE	Low-pass, high-pass, band-pass.	LP / HP / BP
KNOB 4 POSITION	PRE FX: the tracks' dry mix before the effects return (reverb and echoes stay bright). POST REVERB: tracks and effects together, before DRIVE and TAPE. END: the very end, after TAPE and PERFORM.	PRE FX / POST REVERB / END

> The picture: an aurora curtain cut by the filter, low on the left, high on the right.

DRIVE

Tap FX six times



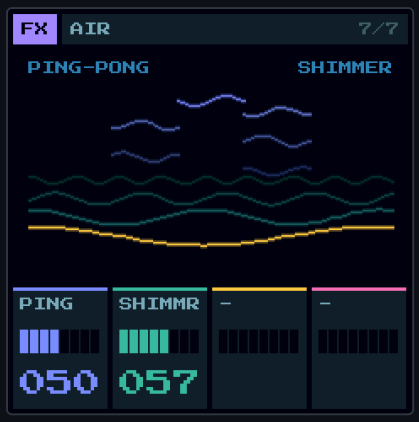
Control	What it does	Range
KNOB 1 AMOUNT	How hard it tears the whole mix, after SPACE, before TAPE. 0: off completely.	OFF, 1–100 %
KNOB 2 TYPE	SOFT: warm saturation. FUZZ: lopsided, buzzy. CLIP: hard. FOLD: a wavefolder, folding back into new overtones.	SOFT / FUZZ / CLIP / FOLD
KNOB 3 TONE	Darkens it.	0–100 %, OPEN
KNOB 4 MIX	Blends it with the clean sound.	0–100 %

> All four types are about equally loud at the same AMOUNT; all the way up is at most ~6 dB louder, and a limiter after it keeps the output safe. Turn AMOUNT on KNOB 1 for fast changes.

> MASTER and DRIVE are saved with the project, like the other effects.

AIR

Tap FX seven times



Control	What it does	Range
KNOB 1 PING	PING-PONG: DELAY's echoes thrown left, right, left, wider as you turn it. While on, TIME goes up to 1.3 s.	OFF, 1–100 %
KNOB 2 SHIMMR	SHIMMER: SPACE's tail taken an octave up and back into the reverb, so it rises as it fades.	OFF, 1–100 %

The picture: DELAY's echoes flying apart above, SHIMMER's octaves rising over SPACE's tail below.

Saved with the project; older projects load with both off.

10

Perform and GENERATE: hold FX

Hold FX for a moment and PERFORM appears. Each trick lasts while you hold its white key; the sequencer keeps running, and you land back in time when you let go. The black keys start GENERATE.

PERFORM

Hold FX + white keys



F2	G2	A2	B2	C3	D3	E3	F3
FREEZE	REV THROW	ECHO THROW	TAPE STOP	REVERSE	FILTER DOWN	FILTER UP	STUTTER

FREEZE is a toggle: tap it to hold the grain cloud, tap again to let go.

GENERATE

Hold FX + black keys



Control	What it does	Range
F#2 G#2 A#2 C#3	Tapped: GENERATE on or off for T1–T4. The track plays by itself: random notes of the SCALE (chord mode: chords), on the tempo, resting now and then.	
D#3	MODE. DRIFT: few long, soft notes. PULSE: a melody on eighths. STORM: dense sixteenths, wild volume.	DRIFT / PULSE / STORM
F#3 / G#3	OCTAVE − / +: the generator's own (0: from C3).	−3 … +3
A#3	RANGE: how far up it may play.	1, 2, 3 OCT

F#2	G#2	A#2	C#3	D#3	F#3	G#3	A#3
T1	T2	T3	T4	MODE	OCT −	OCT +	RANGE

With a track's key held, D#3–A#3 set only that track; alone, all four.

The PERFORM screen shows a tile a track: lit while it generates, its MODE, its range, and a square that flashes with each note. Its own pattern runs on silently underneath and comes back when you switch GENERATE off. Nothing is saved: after power-on it is off.

11

System: hold HOME

Hold HOME for SYSTEM; press HOME again for SYSTEM 2, again for SYSTEM 3. These settings belong to the FM-1, not the project.

SYSTEM

Hold HOME



Control	What it does	Range
KNOB 1 COLOUR	Pick the one that matches your FM-1. MONO NEW: the whole screen black and white, its greys as fine dots.	ICE, AURORA, VIOLET, PLASMA, MONO
KNOB 2 MIDI CHANNEL	1-4: T1–T4 on channels 1–4 (or higher, to 13-16). ALL > SEL: everything to the selected track. CH n > SEL: only that channel, to the selected track.	1-4 … 13-16, ALL, CH 1–16
KNOB 3 TUNE	Fine tuning.	±50 cents
KNOB 4 SCREEN OFF	The screen goes dark after this; the music goes on. The first touch only wakes it. GUOVSSAHAS NEW (the northern lights): after 2 minutes untouched, the aurora over the tundra instead, brighter as the sound grows.	NEVER, 1–30 MIN, GUOVSSAHAS

MONO NEW

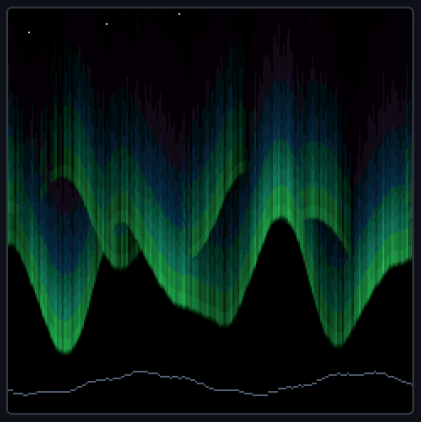
Hold HOME, KNOB 1 to MONO



> The idea after Inigo Hernandez's MONO2K. The header stays clean, so the sound's name reads.

GUOVSSAHAS NEW

Hold HOME, KNOB 4 to GUOVSSAHAS



> Two aurora curtains wave across a night sky, green at their edge, teal and violet above. The first touch brings the page back.

SYSTEM 2

Hold HOME, then HOME



Control	What it does	Range
KNOB 1 MIDI LEARN	Set LEARN, turn a knob on the FM-1, then move one on your controller. Up to 32 links.	OFF, LEARN, CLEAR
KNOB 2 SEQ MIDI OUT	Sends the sequencer's and the arpeggiator's notes out over USB.	OFF / ON
KNOB 3 SAMPLES	CLEAR and OCT+ empties all four sample slots.	KEEP / CLEAR
KNOB 4 CLOCK	INT: its own tempo. I+OUT: the same, and MIDI clock out over USB. USB or TRS: follows a MIDI clock from there (START, CONTINUE, STOP).	INT, I+OUT, USB, TRS

> The processor load, NOW and PEAK, and USB REC DROPS: gaps in the sound sent to a computer. OCT– + OCT+ sets them to 0.

SYSTEM 3

Hold HOME, then HOME twice



Control	What it does	Range
KNOB 1 LATCH	FROM SOUND: a sound you load brings its own LATCH (the seven Holvut sounds latch). ALWAYS OFF: loading sets it off; EDIT's LATCH and SEL + OCT+ still switch it.	FROM SOUND / ALWAYS OFF
KNOB 2 USB LEVEL	How loud a computer records the FM-1. 0 DB: the volume knob all the way up; higher makes quiet drones louder in the recording, a limiter keeps them from clipping.	–12 … +24 DB
KNOB 3 MOD NEW	MOD OUT: each track's level (CC 102) and LFO (CC 103) out over USB 50 times a second, on the track's channel. For lights, motors, Max or another synth.	OFF / ON

MIDI, USB audio and good to know

Control	What it does
MIDI in	USB (the MIDI port is called **Arctic Scream** too), or the TRS jack with a TRS-to-DIN adapter (type A). Notes play the tracks, a CC its MIDI LEARN link; ALL NOTES OFF lets go of MIDI's notes; aftertouch pushes Holvut's FEEDBACK.
USB audio	Class compliant, no driver: the FM-1 shows up on the computer, phone or tablet as **Arctic Scream**, an audio input, 16-bit stereo at 44.1 or 48 kHz, the whole mix at SYSTEM 3's USB LEVEL (the volume knob sets the speaker, not the recording). Solo each track (GLO + keys 5–8) to record them one by one.
Headphones	On wide, loud sound the FM-1's headphone amplifier used to crackle; a limiter now holds the difference between left and right in check near MASTER's top. Mono sound and the USB recording are untouched.
CPU FULL: NOTE DROPPED	It had more to play than it can: a fading or generated note goes first, your held notes last.
WATCHDOG / LOW VOLTAGE RESET	Why it restarted: it stood still, or the power dipped.
Screen reader NEW	The FM-1 says its screen in words: open **charlesvestal.github.io/fm1-reader** in Chrome or Edge and connect. Pages, knobs as they turn, lists, steps and messages, in its own voice or through your screen reader.
Saves NEW	A slot or a sound you save is written at once, a drone sounding or not; the automatic save goes through after 30 s left alone under a latched drone. The editor keeps a project's effects and tempo in its file.
Updating NEW	While a new firmware installs, the screen shows UPDATE MODE, then WRITING with a bar and the per cent. Keep USB in and the FM-1 on until it says DONE.
Newer FM-1s NEW	Units sold since autumn 2026 carry another flash chip (UC25HQ80). Since 1.5.1 saves and updates work on it as on the first one. With 1.5 or older on such a unit, install 1.5.1 once through the chip's download mode (the Firmware Hub's Rescue page).
Saves	Under a never-ending delay, SPACE on INF or a frozen GRAIN, saves go ahead 10 s after the last note. While a computer records over USB, it saves only after a minute left alone.

12

Factory sounds and combos

44 sounds with arctic names. Turn PRESETS to step through them by category, then your own.

Factory sounds

PRESETS

Control	What it does	Range
DRONE	Permafrost, Deep Fjord, Cracking Ice, Arctic Hum, Solar Wind, Under the Ice	
DRONE · Holvut	Polar Howl, Wendigo, Frozen Scream, Whiteout, Dead Signal, Ill, Ice Wraith	latched, DRONE on
PAD	Polar Night, Snow Veil, Glacier Pad, Midnight Sun, Lichen, Snowdrift	
BELL	Ice Bell, Iceberg Gong, Crystal Harp, Bowed Icicle	
VOICE	Aurora Choir, Tundra Voices, Blizzard Breath, Northern Call, Empty Horizon	
KEYS	Frost Keys, Polar Keys, Snow Owl, Signal Flare, Icicle Pluck, Frost Lute, Northern Light	
OTHER	Tundra Strings, Icebreaker, Fog Horn, Svalbard Cello, North Strings	
TEXTURE	Wind over Ice, Biegga	

Combos at a glance

Hold + press

Control	What it does
EDIT + OCT– / OCT+	Undo / redo
FX + white keys	PERFORM: FREEZE … STUTTER
FX + black keys	GENERATE: T1–T4, MODE, OCTAVE, RANGE
SEL (alone)	Chord mode on / off
SEL + a knob	Lock the value to the step (sequencer pages)
SEL + OCT–	Clear the step's locks (sequencer pages)
SEL + OCT+	The track's LATCH on / off
SEQ + white keys	Sequencer tools
SEQ + SELECT	Pattern length
SAVE + white keys	1–4 recall, 5–8 save slots A–D
GLO + white keys	1–4 mute, 5–8 solo T1–T4
ARP (alone)	Arpeggiator on / off
REC (~1 s)	Clear the track's pattern
HOME (alone)	SYSTEM
OCT– + OCT+	Reset CPU PEAK and USB REC DROPS

Arctic Scream p0larpatch by PolarPatch · the demo, the editor and the cheat sheet: polarpatch.github.io/Arctic-scream-FM-1 · credits and licences

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